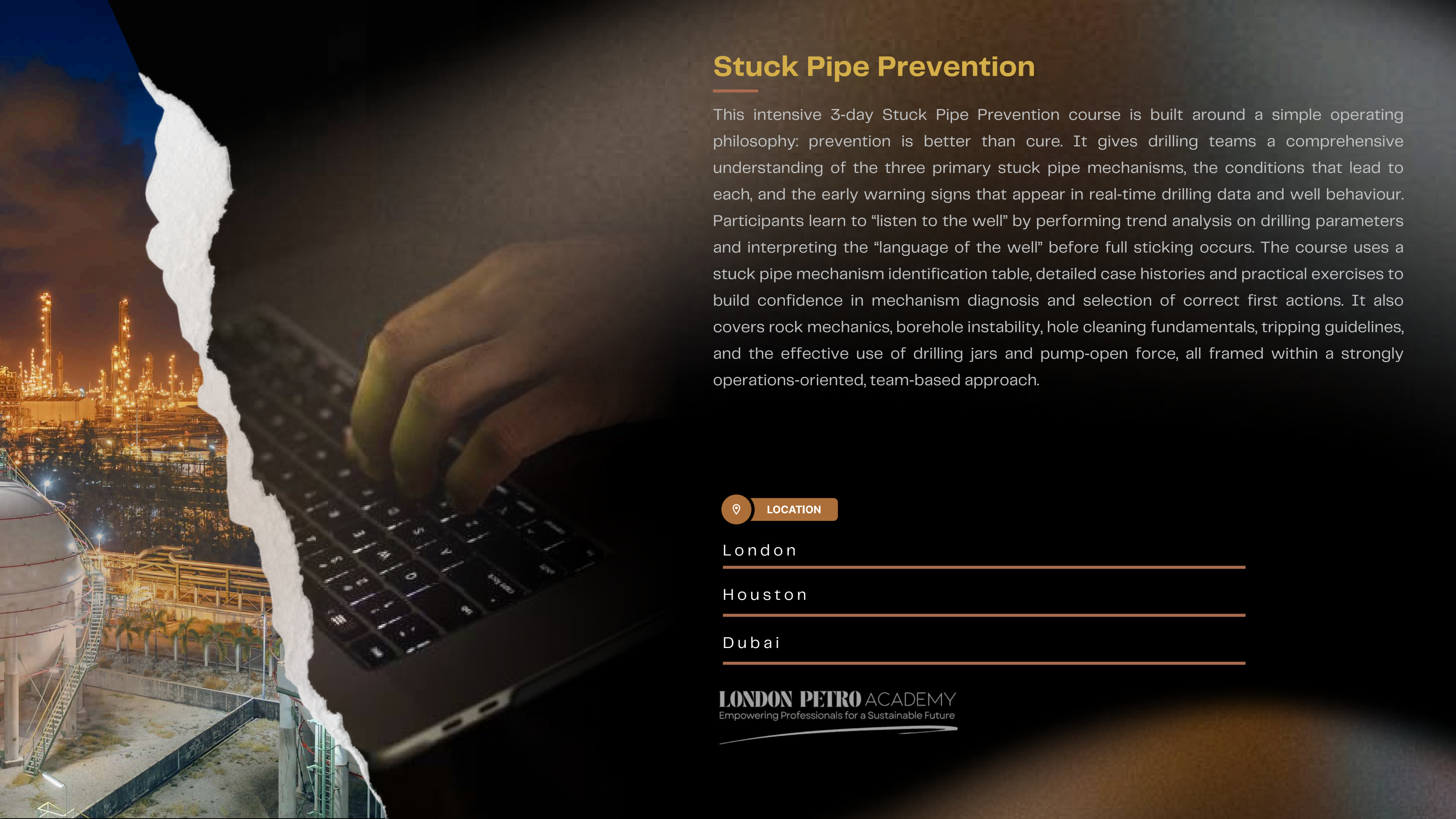




Stuck Pipe Prevention

Proactive Diagnosis, First Actions and Team-Based Practices to Eliminate Stuck Pipe
Non-Productive Time



Stuck Pipe Prevention

This intensive 3-day Stuck Pipe Prevention course is built around a simple operating philosophy: prevention is better than cure. It gives drilling teams a comprehensive understanding of the three primary stuck pipe mechanisms, the conditions that lead to each, and the early warning signs that appear in real-time drilling data and well behaviour. Participants learn to “listen to the well” by performing trend analysis on drilling parameters and interpreting the “language of the well” before full sticking occurs. The course uses a stuck pipe mechanism identification table, detailed case histories and practical exercises to build confidence in mechanism diagnosis and selection of correct first actions. It also covers rock mechanics, borehole instability, hole cleaning fundamentals, tripping guidelines, and the effective use of drilling jars and pump-open force, all framed within a strongly operations-oriented, team-based approach.



LOCATION

London

Houston

Dubai

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



COURSE OBJECTIVES

By the end of this course, participants will be able to identify the three main sticking mechanisms and explain their underlying causes, from hole pack-off and wellbore geometry issues to differential sticking. They will be able to recognise warning signs at an early stage using drilling parameter trends, stratigraphic information and wellbore stress concepts, and select appropriate preventive actions before full sticking develops. Attendees will understand hole cleaning fundamentals, causes of borehole instability and best-practice tripping guidelines, and will have practised using a stuck pipe identification table to link symptoms, mechanisms and recommended first actions. They will also understand how drilling jars work, how to calculate pump-open force and jarring loads, and why jars sometimes fail to free the pipe. Importantly, participants will appreciate the value of a team approach, improved rig-site communication and disciplined first actions, and how these factors can free a high proportion of stuck pipe incidents within the critical first few hours.



This course is designed for rig-site and office-based drilling personnel who are directly involved in planning, supervising or executing drilling operations where stuck pipe risk is a concern. It is particularly suitable for drilling crews, rig managers, drilling foremen, drilling engineers and mud engineers who need a deeper operational understanding of stuck pipe mechanisms, prevention techniques and first-response actions. Wellsite geologists, directional drillers and drilling contractor staff will also benefit from the structured methodology, shared language and team-based diagnostics developed in the course. It is especially effective when taken by entire rig teams as a combined technical and team-building exercise to strengthen communication, shared situational awareness and consistency of response to stuck pipe incidents.

Day 1

Introduction to Oil & Gas Drilling

- Oil and gas development
- Formations, rock pressure, fluid traps
- Fundamentals of petroleum
- Basic geology

Major Causes of Lost Time

- Definitions
- Global statistics
- Stuck pipe causes and mechanisms

Rock Mechanics

- Stratigraphic column
- Deviated and horizontal wells
- Wellbore stresses and instability
- Drilling window
- Drilling fluid properties and mud weight
- Video

Identification of Sticking Mechanisms

- Use of identification table
- Hole pack-off / bridging
- Wellbore geometry
- Differential sticking
- “First Actions”
- Exercise



Day 2

Drilling Jars

- Types of drilling jars
- How jars work
- Pump-open force
- Calculation of jarring loads
- Why jars don't work
- Jars placement
- Exercise

Hole Pack-Off / Bridging

- First mechanism
- Causes
- Drilled cuttings – bit
- Caving – not from bit
- Causes of insufficient hole cleaning
- Annular velocity
- Hole angle
- Boycott effect
- Pipe rotation
- Video
- Hole cleaning guidelines
- “First Actions”
- Exercise
- Causes of bore hole instability
- Tripping guidelines

Well is Talking

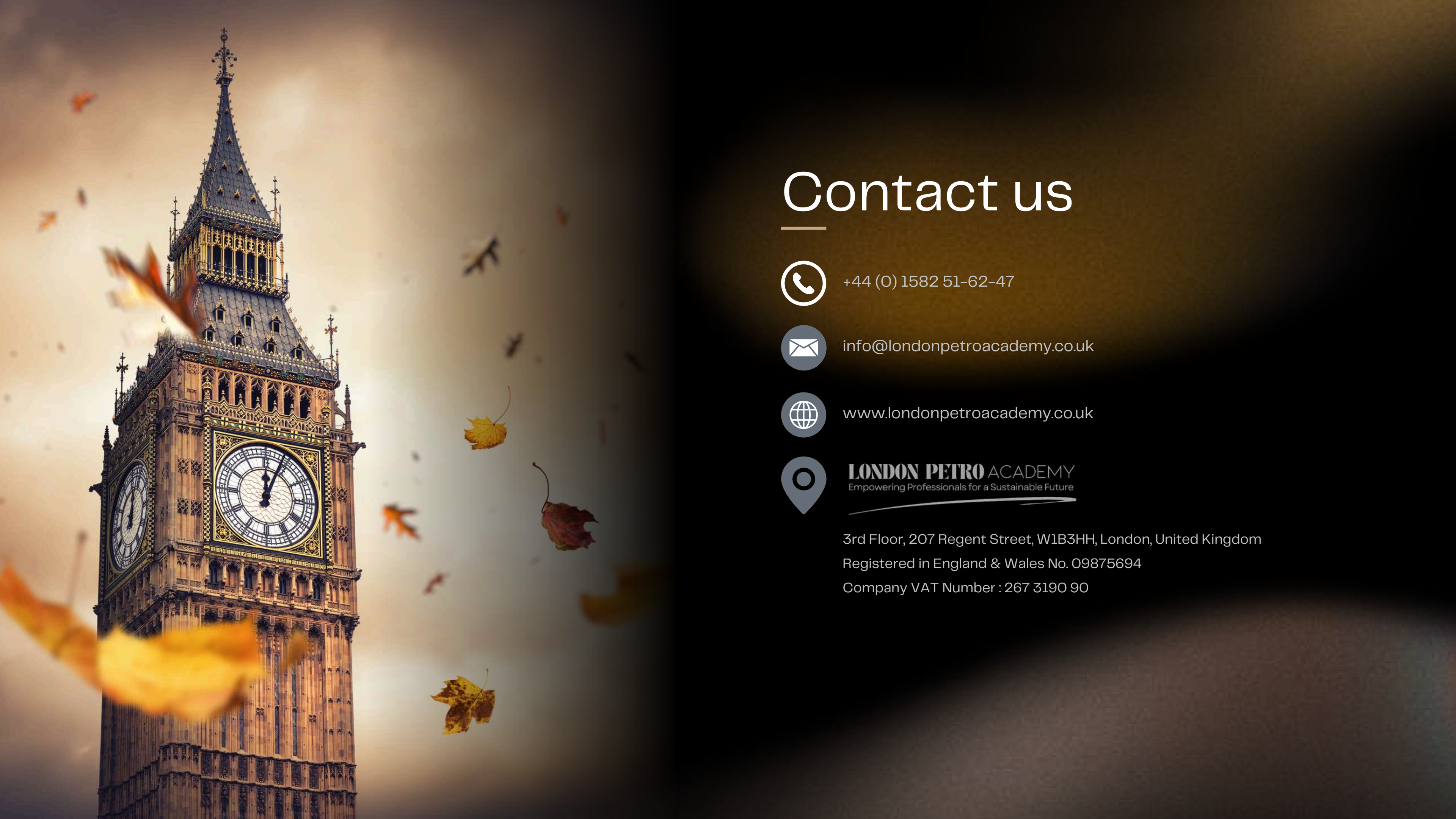
- Drilling parameters recording
- “Learning to Listen” – trend analysis
- Geolograph exercise



Day 3

- Wellbore Geometry
- Second mechanism
- Causes
- Key seating
- Ledges and doglegs
- Under-gauge hole
- Collapsed casing
- Junk
- Shoe joint back-off
- “First Actions” FIRST ACTIONS
- Exercise
- Differential Sticking
- Third mechanism
- Fire prevention triangle and differential sticking prevention square
- Calculation of differential sticking force
- Causes
- Preventive measures
- “First Actions”
- Exercise
- Economics of Fishing
- Time value of money
- Probability model
- Time limit for fishing operations
- Stuck point determination, pipe stretch method
- Consolidation
- Team work
- Implementation of learning
- Recommendations





Contact us



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